

9/11 ch.2.2

Reals: A number with whole part including fractional part

$$R = \pm \left(\overbrace{\sum_{k=1}^n s_k \times b^{k-1} + \dots + s_0 \times b^0}^{\text{integral part}} . \overbrace{\sum_{i=1}^m s_i \times b^{-1} \dots + s_l \times b^{-l}}^{\text{fractional part}} \right)$$

ex. In decimal system: 32.47

$$\begin{array}{ccccccc} & 3 & & 2 & . & 4 & 7 \\ & \times 10^1 & & \times 10^0 & & \times 10^{-1} & \times 10^{-2} \\ 30 & + & 20 & + & 04 & + & 07 & = 32.47 \end{array}$$

Maximum / Minimum Value

$$N_{\max}: 10^k - 1 \quad / \quad N_{\min}: -10^k - 1$$

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Binary system (Base 2): use only two symbols: 0, 1

Binary digits / bits: symbols of binary system.

Integer

Figure 2.2 Place values in an integer in the binary system

	2^{k-1}	2^{k-2}	$\dots$	2^2	2^1	2^0	Place values
$\pm$	S_{k-1}	S_{k-2}	$\dots$	S_2	S_1	S_0	Number
	$\downarrow$	$\downarrow$		$\downarrow$	$\downarrow$	$\downarrow$	
	$N = \pm S_{k-1} \times 2^{k-1}$	$+ S_{k-2} \times 2^{k-2}$	$+ \dots$	$+ S_2 \times 2^2$	$+ S_1 \times 2^1$	$+ S_0 \times 2^0$	Values

ex. 0111

$$\begin{array}{ccccccc} & 0 & & 1 & & 1 & & 1 \\ & \times 2^3 & & \times 2^2 & & \times 2^1 & & \times 2^0 \\ 0 & + & 4 & + & 2 & + & 1 & = 7 \end{array}$$

Reals

$$R = \pm \left(\text{Integral part} \right) + \left(\text{Fractional part} \right)$$

$$R = \pm \left(S_{k-1} \times 2^{k-1} \times \dots \times S_1 \times 2^1 \times S_0 \times 2^0 \right) + \left(S_{-1} \times 2^{-1} + \dots + S_{-l} \times 2^{-l} \right)$$

Hexadecimal system: